

Functions

A function is a machine: put a number in, get one out. Composite = one machine after another (inside first). Inverse = run the machine backwards.



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THE TOPIC ON ONE MAP

Functions

01 — Substitute, domain and range

find $f(10)$

excluded value

range of $(x - 4)^2$

02 — Composite functions

$fg(3)$

$gf(x)$ simplified

solve $fg(x) = 2$

03 — Inverse: linear and fractions

$f^{-1}(x)$ of $\frac{3x+1}{x-2}$

$g^{-1}(x) = 2x$

04 — Inverse of a quadratic

complete the square

domain $x \leq 4$

choose $\pm$

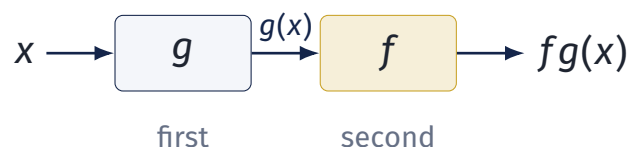
05 — Machines and graphs

function machine

read $g(0)$ from a graph

draw h^{-1}

ORDER MATTERS $fg(x)$ is read right to left — g acts first:



01 Substitute, Domain and Range

$f(10)$: put 10 in place of every x . Domain: the inputs allowed. Range: the outputs you can get. Never divide by zero, never square-root a negative.

WHEN YOU SEE → YOU DO

“find $f(10)$ ”, $f(\frac{1}{2})$

→ replace every x , brackets round negatives

“value of x that cannot be in any domain”

→ bottom = 0: $\frac{11}{2x-5} \rightarrow x = 2.5$

$\sqrt{x-6}$

→ inside ≥ 0 : exclude $x < 6$

“state the range of $f(x) = (x-4)^2$ ”

→ $f(x) \geq 0$

“ $g(a) = -2$, find a ”

→ set the function equal to -2 and solve

ZONE FRACTION

bottom $\neq 0$

$\frac{2x}{x-6}$: exclude $x = 6$

ZONE ROOT

inside ≥ 0

$\sqrt{x-4}$: exclude $x < 4$

WORKED MODEL • SOLVE $g(a) = -2$

$g(x) = \frac{2x}{x+1}$. $g(a) = -2$. Find a .

1 $2a = -2(a+1)$

multiply up

2 $4a = -2$

collect

ANSWER $a = -\frac{1}{2}$

IDEA 01

Practise: Substitute, Domain and Range

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Exclude

CH 3 • TOPIC 3 2 marks

$f(x) = \frac{5}{3x - 6}$. Find $f(4)$ and the value of x that must be excluded.

SELECTED PRACTICE

Q2**PRACTICE** Practice · Medium

CH 3 · TOPIC 3 3 marks

3 (a) f is a function such that

$$f(x) = \frac{1}{x^2 + 1}$$

Find $f\left(\frac{1}{2}\right)$ **(1 mark)****(b)** g is a function such that

$$g(x) = \sqrt{x-1} \quad x \geq 1$$

Find $fg(x)$

Give your answer as simply as possible.

(2 marks)

Q3**PRACTICE**

Practice · Hard

CH 3 · TOPIC 3 5 marks

6 (a) $f(x) = \frac{3}{x+1} + \frac{1}{x-2}$

State one value of x which cannot be included in any domain of f .

(1 mark)

(b) Find the value of $f(0)$

(1 mark)

(c) Find the value of x for which $f(x) = 0$

Show clear algebraic working.

(3 marks)

Q4 **PRACTICE** Practice · Hard

CH 3 · TOPIC 3 3 marks

$$f(x) = cx + d$$

$$23 \quad f(4) = 7$$

$$f(10) = 22$$

Work out the values of c and d .

$$c = \dots\dots\dots$$

$$d = \dots\dots\dots$$

(3 marks)**Q5** **PRACTICE** Practice · Very Hard

CH 3 · TOPIC 3 3 marks

$$14 \quad f(x) = 5 - x \text{ and } g(x) = 3x + 7$$

$$\text{Simplify } f(2x) + g(x - 1)$$

(3 marks)

PAST PAPERS

Q6**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q17

CH 3 • TOPIC 3 4 marks

17 The function f is such that $f(x) = (x - 4)^2$ for all values of x .

(a) Find $f(1)$

.....

(b) State the range of the function f .

.....

The function g is such that $g(x) = \frac{4}{x+3}$ $x \neq -3$

(c) Work out $fg(2)$

.....

(Total for Question 17 is 4 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q14

CH 3 • TOPIC 3 4 marks

14 The function f is defined as

$$f: x \mapsto \frac{2x}{x-6} \quad x \neq 6$$

(a) Find $f(10)$

.....

(b) Express the inverse function f^{-1} in the form $f^{-1}: x \mapsto \dots$

$f^{-1}: x \mapsto$

(Total for Question 14 is 4 marks)

02 Composite Functions

$fg(x)$ means “do g , then do f ”. For a number: inside first. For an expression: put the **whole** of $g(x)$ in place of x in f .

WHEN YOU SEE → YOU DO

“find $fg(3)$ ”

→ $g(3)$ first, then f of that

“find $gf(x)$, simplify”

→ $g(f(x))$: replace x in g by the bracket $f(x)$

“solve $fg(x) = 2$ ”

→ build $fg(x)$, then solve the equation

“ $fg(k) = 2$, work out k ”

→ same: build, equal, solve

“ $ff(x)$ ”

→ f inside itself

Inside function first

Whole expression in brackets

Simplify

Solve if asked

WORKED MODEL • JUNE 2024 P2HR

$f(x) = \frac{x}{2x-4}$, $g(x) = 3x+1$, $fg(k) = 2$. Find k .

$$1 \quad fg(k) = \frac{3k+1}{2(3k+1)-4} = \frac{3k+1}{6k-2}$$

g inside f

$$2 \quad 3k+1 = 12k-4$$

$= 2$, multiply up

ANSWER $k = \frac{5}{9}$

× **TRAP** $fg(x) \neq f(x) \times g(x)$. It is one function inside the other.

IDEA 02

Practise: Composite Functions

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Composite

CH 3 · TOPIC 3 3 marks

$f(x) = 2x + 1$, $g(x) = x^2$. Find $fg(3)$ and $gf(x)$.

SELECTED PRACTICE**Q9** PRACTICE Practice · Medium

CH 3 · TOPIC 3 3 marks

1 (a) f and g are functions such that

$$f(x) = \frac{2}{x^2} \quad \text{and} \quad g(x) = 4x^3$$

Find $f(-5)$

(1 mark)

(b) Find $fg(1)$

(2 marks)

Q10**PRACTICE**

Practice · Hard

CH 3 · TOPIC 3 3 marks

1 $f(x) = 3x^2 - 2x - 8$

Express $f(x + 2)$ in the form $ax^2 + bx$

(3 marks)

Q11**PRACTICE**

Practice · Hard

CH 3 · TOPIC 3 7 marks

12 (a) f is the function such that

$$f(x) = \frac{x}{3x+1}$$

Find $f(0.5)$

(1 mark)

(b) Find $ff(-1)$

(2 marks)

(c) Find the value of x that cannot be included in any domain of f

(1 mark)

(d) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

Show clear algebraic working.

(3 marks)

Q12**PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 3 3 marks

$$18 \quad f(x) = 2x + c$$

$$g(x) = cx + 5$$

$$fg(x) = 6x + d$$

c and d are constants.

Work out the value of d .

(3 marks)

PAST PAPERS

Q13 PAST PAPER Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q16

CH 3 • TOPIC 3 9 marks

16 The functions f and g are defined as

$$f: x \mapsto 5x - 7$$

$$g: x \mapsto \frac{5x}{x + 4}$$

(a) Write down the value of x that must be excluded from any domain of g

.....

(b) Find $gf(2.6)$

.....

(c) Solve $fg(x) = 2$ $x =$ (d) Express the inverse function g^{-1} in the form $g^{-1}: x \mapsto \dots$ $g^{-1}: x \mapsto$

(Total for Question 16 is 9 marks)

Q14**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q17

CH 3 • TOPIC 3 9 marks

17 The functions f and g are defined as

$$f(x) = x^2 + 6$$

$$g(x) = x - 10$$

(a) Find $fg(3)$

.....

(b) Solve the equation $fg(x) = f(x)$

Show clear algebraic working.

.....

The function h is defined as

$$h(x) = \frac{2x - 4}{x}$$

(c) State the value of x that cannot be included in the domain of h

.....

(d) Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots$$

(Total for Question 17 is 9 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q15

CH 3 • TOPIC 3 6 marks

15 The functions f and g are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of x that cannot be included in any domain of g

.....

(b) Find $gf(x)$

Simplify your answer.

$$gf(x) = \text{.....}$$

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$$g^{-1}(x) = \text{.....}$$

(Total for Question 15 is 6 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q19

CH 3 • TOPIC 3 4 marks

19 $f(x) = x^2 - 4$

$g(x) = 2x + 1$

Solve $fg(x) > 0$

Show clear algebraic working.

.....

(Total for Question 19 is 4 marks)

Q17 PAST PAPER Edexcel 4MA1 • May 2023 • Paper 1H • Q17

CH 3 • TOPIC 3 4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \quad x \geq 0$$

(a) What value of x must be excluded from any domain of g ?

.....

(b) Solve $gh(x) = 1$

.....

(Total for Question 17 is 4 marks)

Q18 PAST PAPER Edexcel 4MA1 • June 2024 • Paper 2HR • Q17

CH 3 • TOPIC 3 3 marks

17 $f(x) = \frac{x}{2x - 4}$ $g(x) = 3x + 1$

Given that $fg(k) = 2$

work out the value of k

 $k =$

(Total for Question 17 is 3 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1H · Q16

CH 3 · TOPIC 3 4 marks

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$
 $g(x) = 4x^2$

(a) Find $f(9)$

.....

(b) Solve $fg(x) < 0$
Show clear algebraic working.

.....

(Total for Question 16 is 4 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q19

CH 3 • TOPIC 3 3 marks

19 The functions f and g are such that

$$f(x) = 3x - 4 \text{ where } x > 2$$

$$g(x) = \frac{x}{2x + 1}$$

(a) State the value of x that cannot be included in any domain of g

.....

(b) Find $gf(x)$

Give your answer in its simplest form.

$$gf(x) = \text{.....}$$

(Total for Question 19 is 3 marks)

Q21**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q19

CH 3 • TOPIC 3 3 marks

19 The functions f and g are such that

$$f(x) = 3x - 2$$

$$g(x) = \frac{x}{2x - 1}$$

(a) Find $g(3)$

.....

(b) Find $gf(x)$

Give your answer in its simplest form.

$$gf(x) = \text{.....}$$

(Total for Question 19 is 3 marks)

03 Inverse: Linear and Fractions

Write $y = f(x)$, make x the subject (Topic 2.7), then swap x and y . With x twice in a fraction: multiply up, collect x , factorise.

WHEN YOU SEE → YOU DO

“express f^{-1} in the form $f^{-1}(x) = \dots$ ”

→ $y = f(x) \rightarrow x = \dots \rightarrow$ swap

$\frac{3x+1}{x-2}$ (x on top and bottom)

→ multiply up → collect x → factorise

“solve $g^{-1}(x) = 2x$ ”

→ find g^{-1} first, then solve

“ $f^{-1}(55) = fg(4)$, show that”

→ work out each side separately

$y = f(x)$

Make x the subject

Swap x and y

WORKED MODEL • MAY 2024 P1H

$f(x) = \frac{3x+1}{x-2}$. Find $f^{-1}(x)$.

1 $y(x-2) = 3x+1$

multiply up

2 $xy - 3x = 2y + 1 \Rightarrow x(y-3) = 2y+1$

collect, factorise

ANSWER $f^{-1}(x) = \frac{2x+1}{x-3}$

IDEA 03

Practise: Inverse: Linear and Fractions

My questions first, then selected practice, then real past papers.

Q22 DR ESLAM Inverse

CH 3 · TOPIC 3 3 marks

$$f(x) = \frac{x+4}{x-1}. \text{ Find } f^{-1}(x).$$

SELECTED PRACTICE**Q23** PRACTICE Practice · Medium

CH 3 · TOPIC 3 1 mark

$$11 \quad f(x) = 3x$$

Circle the expression for $f^{-1}(x)$

$-3x$	$\frac{3}{x}$	$\frac{1}{3x}$	$\frac{x}{3}$
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(1 mark)

Q24**PRACTICE**

Practice · Hard

CH 3 · TOPIC 3 4 marks

14 Two functions, f and g are defined as

$$f : x \mapsto 1 + \frac{1}{x} \text{ for } x > 0$$

$$g : x \mapsto \frac{x+1}{2} \text{ for } x > 0$$

Given that $h = fg$

express the inverse function h^{-1} in the form $h^{-1} : x \mapsto \dots$

$$h^{-1} : x \mapsto \dots$$

(4 marks)

Q25**PRACTICE**

Practice · Hard

CH 3 · TOPIC 3 3 marks

21 $g(x) = 3x + 7$

Solve $g^{-1}(x) = 2x$

$x = \dots\dots\dots$

(3 marks)**Q26****PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 3 5 marks

13 $f(x) = \frac{2x}{5} - 1$

Work out the value of $f^{-1}(3) + f(-0.5)$

(5 marks)

PAST PAPERS

Q27**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q15

CH 3 • TOPIC 3 4 marks

15 The function f is defined as

$$f: x \mapsto \frac{3x + 1}{x - 2}$$

(a) State the value of x that cannot be included in any domain of the function f

.....

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$

(Total for Question 15 is 4 marks)

04 Inverse of a Quadratic

A quadratic has x in two places. **Complete the square first** (Dr Eslam's rule), then Move $\rightarrow$ Root $\rightarrow$ Solve. The domain tells you $+\sqrt{\quad}$ or $-\sqrt{\quad}$.

WHEN YOU SEE $\rightarrow$ YOU DO

$f(x) = x^2 - 8x + 5$ where $x \leq 4$, "find f^{-1} " $\rightarrow (x - 4)^2 - 11$, then $\pm$

domain $x \geq$ the vertex $\rightarrow$ take $+\sqrt{\quad}$

domain $x \leq$ the vertex $\rightarrow$ take $-\sqrt{\quad}$

"state the domain of g^{-1} " $\rightarrow$ domain of g^{-1} = range of g

Complete the square

Move

Square root (choose the sign)

Swap

WORKED MODEL • JANUARY 2021 P1H

$f(x) = x^2 - 8x + 5$, $x \leq 4$. Find $f^{-1}(x)$.

1 $y = (x - 4)^2 - 11$

complete the square

2 $(x - 4)^2 = y + 11 \Rightarrow x - 4 = -\sqrt{y + 11}$

$x \leq 4$: negative root

ANSWER $f^{-1}(x) = 4 - \sqrt{x + 11}$

× **TRAP** Keeping $\pm$ in the final answer. A function gives one output — the domain chooses the sign.

IDEA 04

Practise: Inverse of a Quadratic

My questions first, then selected practice, then real past papers.

Q28 **DR ESLAM** Complete the square

CH 3 · TOPIC 3 4 marks

$f(x) = x^2 + 6x + 2$, $x \geq -3$. Find $f^{-1}(x)$.

SELECTED PRACTICE**Q29** **PRACTICE** Practice · Very Hard

CH 3 · TOPIC 3 7 marks

9 (a) The function f is such that $f(x) = 5 + 6x - x^2$ for $x \leq 3$

Express $5 + 6x - x^2$ in the form $p - (x - q)^2$ where p and q are constants.

(2 marks)

(b) Using your answer to part (a), find the range of values of x for which $f^{-1}(x)$ is positive.

(5 marks)

Q30**PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 3 5 marks

10 (a) g is the function with domain $x \geq -3$ such that $g(x) = x^2 + 6x$

Write down the range of g^{-1}

(1 mark)

(b) Express the inverse function g^{-1} in the form $g^{-1} : x \mapsto \dots$

 $g^{-1} : x \mapsto$ **(4 marks)**

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1H • Q21

CH 3 • TOPIC 3 5 marks

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geq -2$ Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

Q32 PAST PAPER Edexcel 4MA1 • January 2021 • Paper 1H • Q22

CH 3 • TOPIC 3 3 marks

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leq 4$ Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots\dots\dots$$

(Total for Question 22 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 1H · Q24

CH 3 · TOPIC 3 6 marks

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

.....

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots$$

(Total for Question 24 is 6 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • January 2022 • Paper 2HR • Q23**CH 3 • TOPIC 3** **4 marks**

23 The functions f and g are such that

$$f(x) = x + 25 \quad g(x) = x^2 - 12x$$

The function h is such that $h(x) = fg(x)$

The domain of h is $\{x : x \leq 6\}$

Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

Q35**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q25

CH 3 • TOPIC 3 5 marks

25 The function g is defined as

$$g:x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x:x \geq 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1}:x \mapsto \dots$

$$g^{-1}:x \mapsto \dots$$

(b) State the domain of g^{-1}

.....

(Total for Question 25 is 5 marks)

Q36 **PAST PAPER** Edexcel 4MA1 · November 2023 · Paper 2H · Q19**CH 3 · TOPIC 3** **7 marks**

19 The functions f and g are such that

$$f:x \mapsto 5x + 7$$

$$g:x \mapsto \frac{5}{2x - 9}$$

(a) State which value of x cannot be included in any domain of g

.....

(b) Find $fg(4)$

.....

The function h is such that

$$h:x \mapsto 3x^2 - 12x + 8 \quad \text{where } x > 2$$

(c) Express the inverse function h^{-1} in the form $h^{-1}:x \mapsto \dots$

$$h^{-1}:x \mapsto \dots$$

(Total for Question 19 is 7 marks)

Q37**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q25

CH 3 • TOPIC 3 4 marks

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geq 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

05 Machines and Graphs

A function machine is a function in boxes. A graph gives $f(x)$: go up from x , read across. For f^{-1} reflect in $y = x$.

WHEN YOU SEE → YOU DO

“ $\times 2$ then $+5$, the second output is 11”

→ work backwards through both runs

“find $g(0)$ ” from a graph

→ go up from $x = 0$, read y

“find $gf(-1)$ ” from two graphs

→ read $f(-1)$, then read g of that

“draw $y = h^{-1}(x)$ ”

→ reflect $y = h(x)$ in the line $y = x$

“ $f(x) = k$ has only one solution”

→ horizontal line $y = k$ meets the curve once

WORKED MODEL • MACHINE

Machine: $\times 2$ then $+5$. The output is used as a new input; the second output is 11. Find the first input.

1 second run: $(11 - 5) \div 2 = 3$

backwards

2 first run: $(3 - 5) \div 2$

backwards again

ANSWER -1

IDEA 05

Practise: Machines and Graphs

My questions first, then selected practice.

SELECTED PRACTICE

Q38

PRACTICE

Practice · Medium

CH 3 · TOPIC 3 5 marks

12 (a) A function is represented by the following function machine.



A number is input into the machine.

The output is used as a new input.

The second output is 11.

Work out the number that was the **first input**.

(2 marks)

(b) A number is input into the machine.

The output given is the same number.

Work out the number.

(3 marks)

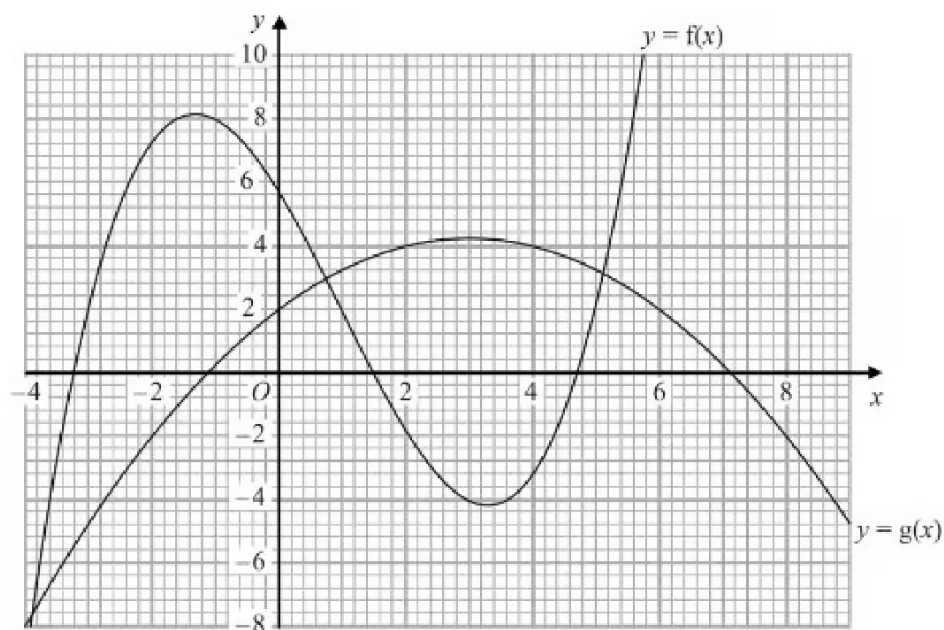
Q39**PRACTICE**

Practice · Hard

CH 3 · TOPIC 3

3 marks

7 (a) The diagram shows parts of the graphs of $y = f(x)$ and $y = g(x)$.

Find $g(0)$ **(1 mark)**(b) Find $gf(-1)$ **(2 marks)**

Q40**PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 3 9 marks

3 (a)

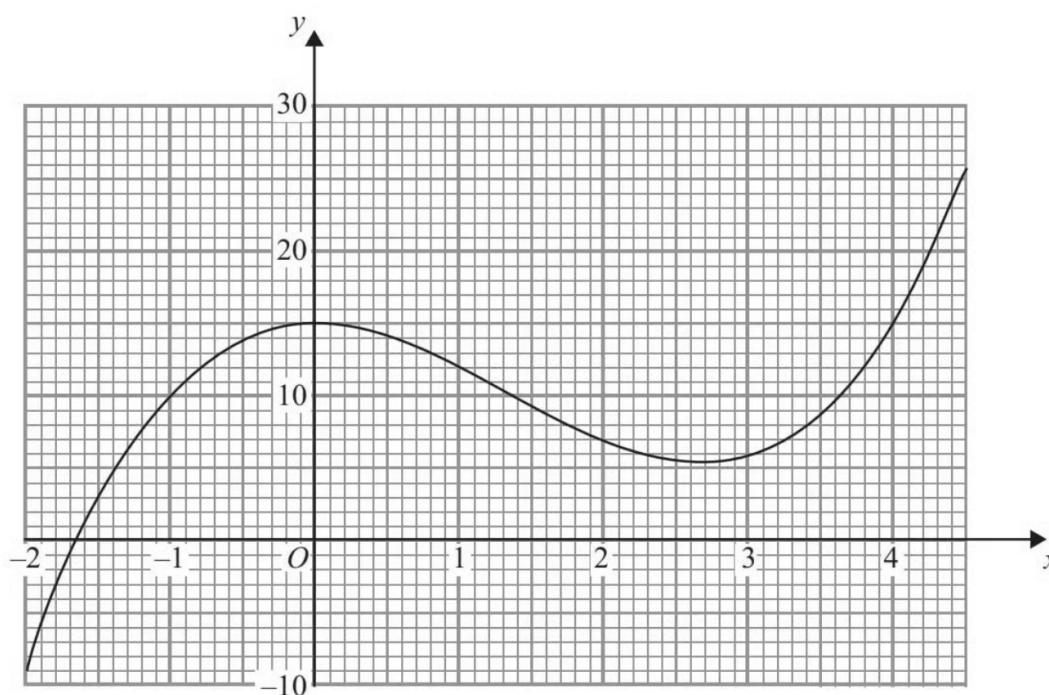
$$f(x) = \sqrt{x-6}$$

Find $f(10)$

(1 mark)

(b) State which values of x must be excluded from a domain of f

(2 marks)

(c) The diagram shows part of the graph of $y = g(x)$ Find $g(2)$

(1 mark)

(d) Find $fg(0)$

(2 marks)

(e) One of the solutions of $g(x) = k$, where k is a number, is $x = 1$

Find the other solutions

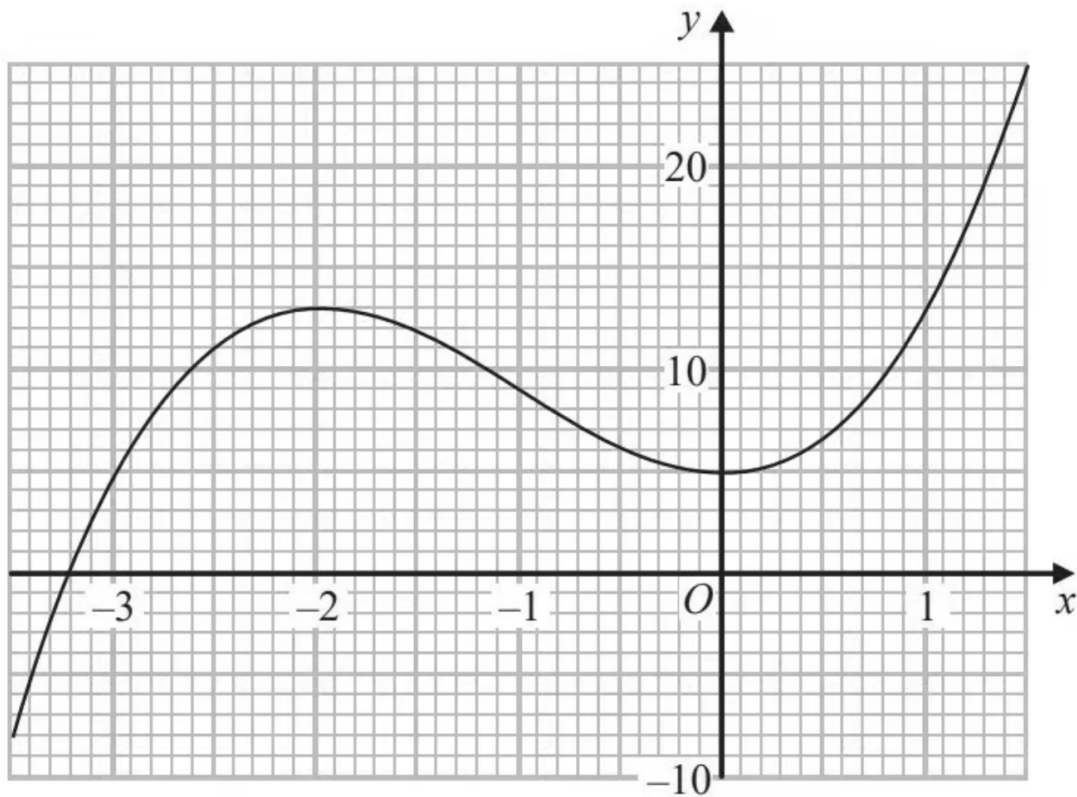
Give your answers correct to 1 decimal place

(3 marks)

Q41 PRACTICE Practice · Very Hard

CH 3 · TOPIC 3 9 marks

- 4 (a) The diagram shows the graph of $y = f(x)$ for $-3.5 \leq x \leq 1.5$



Use the graph of $f(x)$ to find the value of $f(0)$.

(1 mark)

- (b) For which values of k does the equation $f(x) = k$ have only one solution?

(2 marks)

- (c) Find an estimate for the gradient of the curve at the point where $x = -2.5$

(3 marks)

(d) $g(x) = \frac{1}{2+x}$

State which value of x must be excluded from any domain of g

(1 mark)

- (e) Find $fg(-3)$

(2 marks)

Q42**PRACTICE**

Practice · Very Hard

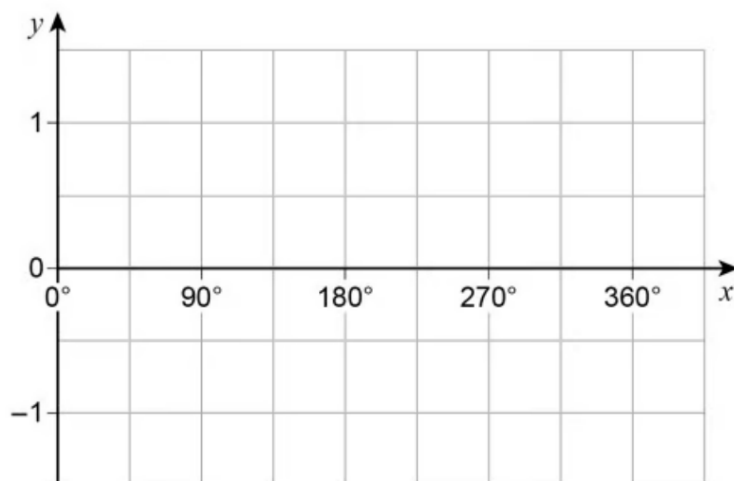
CH 3 · TOPIC 3 2 marks

16 For all values of x

$$f(x) = \sin x$$

$$g(x) = x + 90$$

On the grid, draw the graph of the composite function $y = fg(x)$ for $0^\circ \leq x \leq 360^\circ$

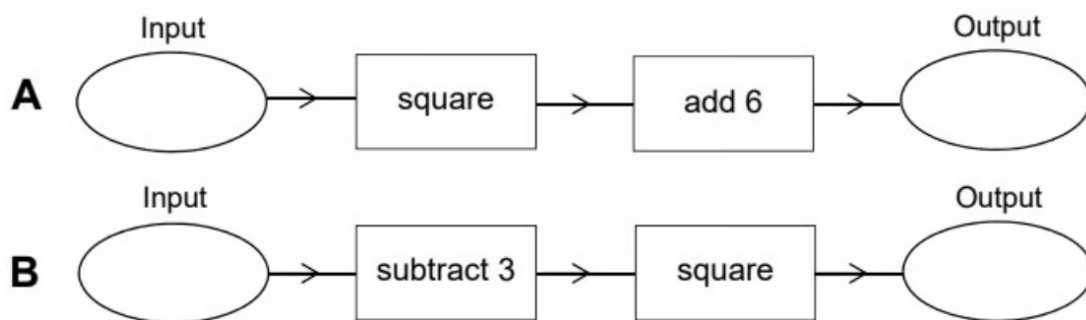
**(2 marks)**

Q43**PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 3 4 marks

17 Here are two function machines, **A** and **B**.



Both machines have the same input.

Work out the range of input values for which

the output of **A** is **less** than the output of **B**.

(4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|---|
| 01 | Domain / range | Bottom $\neq 0$; root ≥ 0 ; range = outputs. |
| 02 | Composite | fg : g first; the whole bracket goes in. |
| 03 | Inverse | $y = f(x)$, make x the subject, swap. |
| 04 | Quadratic inverse | Complete the square; the domain chooses the sign. |
| 05 | Machines / graphs | Backwards through the boxes; f^{-1} reflects in $y = x$. |

BOTTOM LINE fg : inside first $\rightarrow f^{-1}$: rearrange and swap $\rightarrow$ quadratic: complete the square first.

Answers

Functions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SUBSTITUTE, DOMAIN AND RANGE

- | | | |
|-----------|---|--|
| Q1 | Dr Eslam • Exclude | $f(4) = \frac{5}{6}; x = 2$ |
| Q2 | Practice • Medium | (a) $\frac{4}{5}$ (b) $fg(x) = \frac{1}{x}$ |
| Q3 | Practice • Hard | (a) -1 or 2 (b) 2.5 (c) $x = \frac{5}{4}$ |
| Q4 | Practice • Hard | $c = 2.5, d = -3$ |
| Q5 | Practice • Very Hard | $x + 9$ |
| Q6 | Edexcel 4MA1 • January 2020 • Paper 1HR • Q17 | (a) 9 (b) $f(x) \geq 0$ (c) $\frac{256}{25}$ |
| Q7 | Edexcel 4MA1 • June 2022 • Paper 1H • Q14 | (a) 5 (b) $\frac{6x}{x-2}$ |

IDEA 02 • COMPOSITE FUNCTIONS

- Q8** Dr Eslam • Composite $fg(3) = 19; gf(x) = (2x + 1)^2$
- Q9** Practice • Medium (a) $\frac{2}{25}$ (b) $\frac{1}{8}$
- Q10** Practice • Hard $3x^2 + 10x$
- Q11** Practice • Hard (a) 0.2 (b) 0.2 (c) $-\frac{1}{3}$ (d) $\frac{x}{1 - 3x}$
- Q12** Practice • Very Hard $c = 3, d = 13$
- Q13** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q16 (a) -4 (b) 3 (c) $x = \frac{9}{4}$ (d) $\frac{4x}{5 - x}$
- Q14** Edexcel 4MA1 • January 2021 • Paper 2HR • Q17 (a) 55 (b) $x = 5$ (c) 0 (d) $\frac{4}{2 - x}$
- Q15** Edexcel 4MA1 • November 2021 • Paper 2H • Q15 (a) $-\frac{1}{3}$ (b) $\frac{2x - 3}{6x - 8}$ (c) $\frac{x}{1 - 3x}$
- Q16** Edexcel 4MA1 • June 2022 • Paper 2HR • Q19 $x < -1.5$ or $x > 0.5$
- Q17** Edexcel 4MA1 • May 2023 • Paper 1H • Q17 (a) 2.5 (b) $x = 2$
- Q18** Edexcel 4MA1 • June 2024 • Paper 2HR • Q17 $k = \frac{5}{9}$
- Q19** Edexcel 4MA1 • June 2025 • Paper 1H • Q16 (a) 6 (b) $9 - 2|x| < 0: x < -4.5$ or $x > 4.5$
- Q20** Edexcel 4MA1 • June 2025 • Paper 2HR • Q19 (a) $-\frac{1}{2}$ (b) $\frac{3x - 4}{6x - 7}$
- Q21** Edexcel 4MA1 • November 2025 • Paper 2H • Q19 (a) $\frac{3}{5}$ (b) $\frac{3x - 2}{6x - 5}$

IDEA 03 • INVERSE: LINEAR AND FRACTIONS

Q22 Dr Eslam • Inverse

$$xy - y = x + 4: f^{-1}(x) = \frac{x+4}{x-1} \text{ (self-inverse)}$$

Q23 Practice • Medium

$$\frac{x}{3}$$

Q24 Practice • Hard

$$\frac{3-x}{x-1}$$

Q25 Practice • Hard

$$x = -\frac{7}{5}$$

Q26 Practice • Very Hard

$$10 + (-1.2) = 8.8$$

Q27 Edexcel 4MA1 • May 2024 • Paper 1H • Q15

$$(a) 2 \quad (b) \frac{2x+1}{x-3}$$

IDEA 04 • INVERSE OF A QUADRATIC

- Q28** Dr Eslam · Complete the square $(x + 3)^2 - 7: f^{-1}(x) = -3 + \sqrt{x + 7}$
- Q29** Practice · Very Hard (a) $14 - (x - 3)^2$ (b) $5 < x \leq 14$
- Q30** Practice · Very Hard (a) $g^{-1}(x) \geq -3$ (b) $-3 + \sqrt{x + 9}$
- Q31** Edexcel 4MA1 · January 2020 · Paper 1H · Q21 $-2 + \sqrt{x + 1}$
- Q32** Edexcel 4MA1 · January 2021 · Paper 1H · Q22 $4 - \sqrt{x + 11}$
- Q33** Edexcel 4MA1 · May 2021 · Paper 1H · Q24 (a) 247 (b) $1 + \sqrt{\frac{x-2}{5}}$
- Q34** Edexcel 4MA1 · January 2022 · Paper 2HR · Q23 $6 - \sqrt{x + 11}$
- Q35** Edexcel 4MA1 · January 2022 · Paper 2H · Q25 (a) $3 + \sqrt{14 - x}$ (b) $x \leq 14$
- Q36** Edexcel 4MA1 · November 2023 · Paper 2H · Q19 (a) 4.5 (b) -18 (c) $2 + \sqrt{\frac{x+4}{3}}$
- Q37** Edexcel 4MA1 · November 2024 · Paper 1H · Q25 $6 + \sqrt{\frac{x+65}{2}}$

IDEA 05 · MACHINES AND GRAPHS

- Q38** Practice · Medium (a) -1 (b) -5
- Q39** Practice · Hard (a) $g(0) = 2$ (b) $f(-1) = 8$, $g(8) \approx -2$
- Q40** Practice · Very Hard (a) 2 (b) $x < 6$ (c) 7 (d) 3 (e) $x \approx -0.8, 3.8$
- Q41** Practice · Very Hard (a) 5 (b) $k < 5$ or $k > 13$ (c) about 7.5 (d) -2 (e) 9
- Q42** Practice · Very Hard $y = \cos x$ (from 1 at 0° to -1 at 180°)
- Q43** Practice · Very Hard $x^2 + 6 < (x - 3)^2$: $x < \frac{1}{2}$